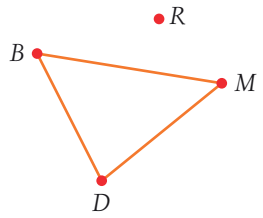
WORKSHEET

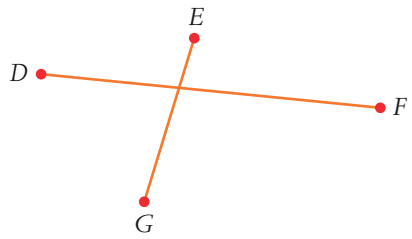
Chapter 5: Adjacency matrices

1 Write the adjacency matrix for each graph.

a

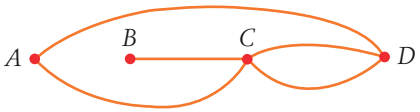


b

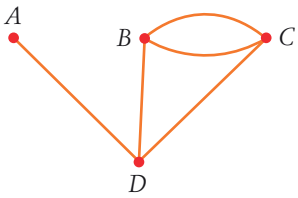


© Cengage Learning Australia Pty Ltd 2014

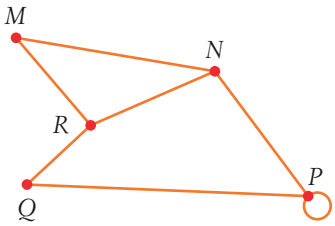
c



d

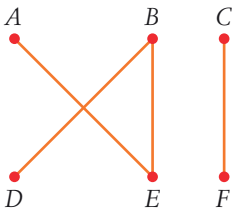


e



© Cengage Learning Australia Pty Ltd 2014

f



2 Draw the graph for each adjacency matrix.

a

$$\begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \end{array}$$

b

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

© Cengage Learning Australia Pty Ltd 2014

c

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{bmatrix} \end{array}$$

d

$$\begin{array}{c} \\ J \\ K \\ L \\ M \end{array} \begin{array}{c} \begin{array}{cccc} J & K & L & M \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

© Cengage Learning Australia Pty Ltd 2014

e

$$\begin{array}{c} \\ V \\ W \\ X \\ Y \\ Z \end{array} \begin{array}{c} \begin{array}{ccccc} V & W & X & Y & Z \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

f

$$\begin{array}{c} \\ M \\ N \\ P \\ Q \\ R \end{array} \begin{array}{c} \begin{array}{ccccc} M & N & P & Q & R \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

© Cengage Learning Australia Pty Ltd 2014

Answers

1 a

	B	D	M	R
B	0	1	1	0
D	1	0	1	0
M	1	1	0	0
R	0	0	0	0

b

	D	E	F	G
D	0	0	1	0
E	0	0	0	1
F	1	0	0	0
G	0	1	0	0

c

	A	B	C	D
A	0	0	1	1
B	0	0	1	0
C	1	1	0	2
D	1	0	2	0

d

	A	B	C	D
A	0	0	0	1
B	0	0	2	1
C	0	2	0	1
D	1	1	1	0

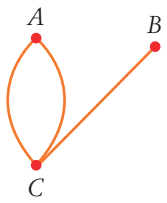
e

	M	N	P	Q	R
M	0	1	0	0	1
N	1	0	1	0	1
P	0	1	1	1	0
Q	0	0	1	0	1
R	1	1	0	1	0

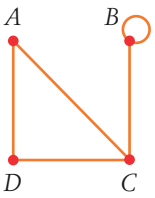
f

	A	B	C	D	E	F
A	0	0	0	0	1	0
B	0	0	0	1	1	0
C	0	0	0	0	0	1
D	0	1	0	0	0	0
E	1	1	0	0	0	0
F	0	0	1	0	0	0

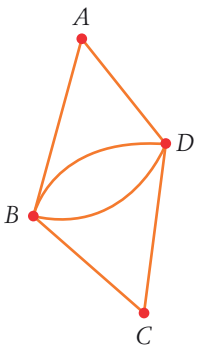
2 a



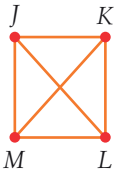
b



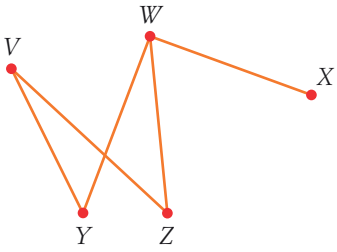
c



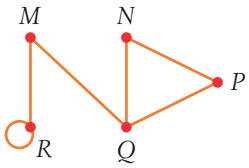
d



e



f



© Cengage Learning Australia Pty Ltd 2014